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NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 2 EXAMINATION 2015-2016

CE2005/CZ2005 – OPERATING SYSTEMS

Apr/May 2016 Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 7 pages.
2. Answer **ALL** questions.
3. This is a closed-book examination.
4. Questions **DO NOT** carry equal marks.

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1. (a) Indicate whether the following statements are true or false. Justify your answers.

(i) Privileged instructions can be executed in user mode.
(2 marks)

(ii) Real-time operating systems usually use preemptive job scheduling.
(2 marks)

(iii) The system handles the input from mouse with interrupt-driven I/O, instead of DMA.
(2 marks)

(iv) A multi-processing system usually has the multi-programming capacity.
(2 marks)

(v) If in a CPU scheduling policy, a process can transit from running state to waiting state, we can conclude that this CPU scheduling policy is preemptive.
(2 marks)

Note: Question No. 1 continues on Page 2

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(b) Suppose two processes P1 and P2 enter the system at time zero in the order of P1 and P2. They have the following actions:

P1 has a single CPU burst of 5 units and does not have any I/O burst. P2 has two occurrences of alternating CPU and I/O bursts. In each occurrence, P2 first performs a CPU burst of 2 units, and then performs an I/O burst of 2 units, i.e., the total elapsed time for P2 is 8 units.

(i) Consider the scenario that P1 is in the ready queue and P2 is in the running state. Briefly explain the operations that the system will perform on the process control blocks (PCB) of P1 and P2, when P2 initiates the I/O burst.

(3 marks)

(ii) Show the average turn-around time for P1 and P2 with each of the following two scheduling algorithms: (a) shortest-job-first (preemptive) and (b) round-robin (quantum = 2 units).

(4 marks)

(c) Jim has written a C/C++ program, as shown in Figure Q1. Consider the system with dual-mode protection.

(i) List one line of source code with mode switches and briefly justify in your answer why mode switches are necessary.

(4 marks)

(ii) Describe how the four memory sections of the process (i.e., text, data, heap and stack sections) change in the entire process execution.

(4 marks)

```
Line 1: void main () {
Line 2:     f1(5);
Line 3:  }
Line 4: void f1(int times) {
Line 5:     if(times>0){
Line 6:         int pid=fork();
Line 7:         printf("%d\n", pid);
Line 8:         f1(times-1);
Line 9:     }
Line 10:    else
Line 11:        printf("done\n");
Line 12: }
```

Figure Q1

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2. (a) Indicate whether the following statements are true or false. Justify your answers.

(i) The system is in safe state if there is a sequence for finishing all processes.

(2 marks)

(ii) Semaphore with blocking is more efficient than that with busy waiting.

(2 marks)

(iii) Simple batch systems do not have deadlock problems.

(2 marks)

(iv) Deadlock will not occur in the Dining-Philosophers problem, if the system preempts a philosopher who cannot get the other chopstick for a long time.

(2 marks)

(v) If the resource allocation graph contains more than one cycle, a deadlock must occur in the system.

(2 marks)

(b) Consider the following snapshot of a system's state, with three processes (P0, P1 and P2) and four resource types (A, B, C and D). The current Allocation and Needed matrices are as shown in Table Q2a. What is the minimum value of x in Available so that the system is deadlock-free?

(4 marks)

Table Q2a

Process	Allocation A B C D	Available x 1 1 0 Needed A B C D
P0	1 1 2 0	1 1 0 2
P1	0 1 1 1	2 0 1 0
P2	0 0 1 1	1 1 0 0

Note: Question No. 2 continues on Page 4

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(c) A molecule of water H_2O consists of two atoms of Hydrogen and one atom of Oxygen. We can model atom productions of Hydrogen and Oxygen as two kinds of co-operating processes, illustrated in Figure Q2. Multiple Hydrogen and Oxygen processes can be created. Each process produces one atom (either a Hydrogen or an Oxygen atom). The Oxygen process will deliver a water molecule after assembling two atoms of Hydrogen and one atom of Oxygen. Use semaphores to implement the synchronization between these two kinds of processes. You are only allowed to define semaphores in A and to use semaphore operations (Wait and Signal) in B/C. Your implementation should be deadlock free.

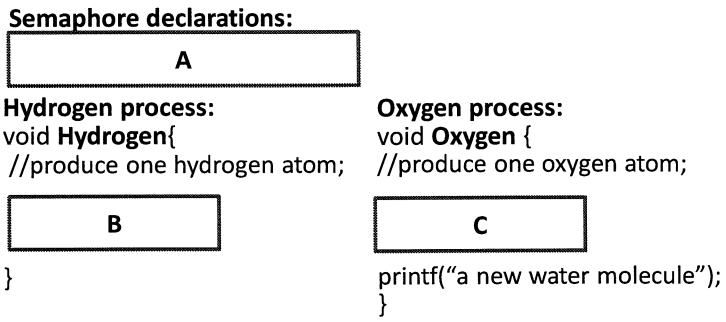


Figure Q2

(6 marks)

(d) Consider two processes P1 and P2 in Table Q2b. Semaphores S and Q are both initialized to be 0. X, Y and Z are shared variables among P1 and P2. X, Y and Z are all initialized with 2. What are all the possible values for X, Y and Z after P1 and P2 complete?

Table Q2b

Process P1	Process P2
Y=1;	X=1;
signal(S);	X=X+1;
Y=Y+Z;	wait(S);
Z=Y+1;	X=X-Y;
wait(Q);	signal(Q);
Y=Z-Y;	Z=X+Z;

(5 marks)

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3. (a) Consider a computer system using dynamic partitioning approach for memory allocation. The memory currently has the following holes of various sizes in physical address order: 4 KB, 8 KB, 20 KB, 9 KB, 14 KB, 10 KB, and 40 KB. Given a request of 12 KB, identify the hole to be allocated for each of the following storage allocation strategies:

- First Fit
- Best Fit
- Worst Fit

(6 marks)

(b) Figure Q3 shows the mapping of logical address of a process to the physical memory. "×" in the figure indicates the corresponding page is not in the physical memory.

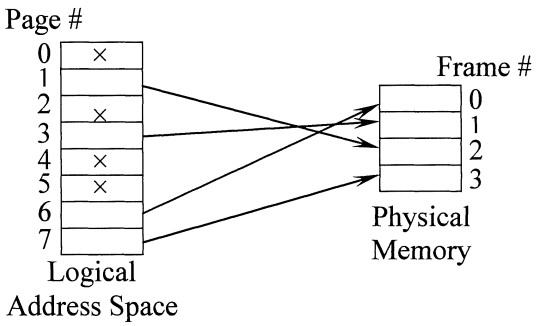


Figure Q3

(i) Draw the page table of the process. Clearly indicate the content of each entry in the page table.

(4 marks)

(ii) Given the following logical addresses, what are the corresponding physical addresses?

- 001101101011
- 010110110101

(4 marks)

(iii) What is the page size?

(2 marks)

(c) Explain what "thrashing" means in the context of memory management and state the cause for such a phenomenon. Describe a strategy to prevent this from occurring.

(5 marks)

Note: Question No. 3 continues on Page 6

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(d) Assume a computer system uses demand paging for virtual memory management. There are four page frames which are initially empty.

(i) For the following page reference sequence, how many page faults will be generated if First-In-First-Out (FIFO) page replacement algorithm is used?

0, 1, 2, 3, 4, 5, 0, 1, 2, 3, 4, 5

(2 marks)

(ii) Will result be different if Least-Recently-Used (LRU), instead of FIFO, page replacement algorithm is used?

(2 marks)

(iii) Design a realistic page replacement algorithm that would perform better than FIFO and LRU for this reference sequence.

(3 marks)

4. (a) Indicate whether the following statements about file system and I/O are true or false. Justify your answers.

(i) Using hard link, rather than symbolic link, a file can have different names.

(2 marks)

(ii) Assuming the protection bits of a file are "r w x r x - r - -", all the users in this Unix-based system are able to read and modify the content of this file.

(2 marks)

(iii) Using link allocation method for file space allocation may make disk fragmented.

(2 marks)

(iv) A process will be in either running or ready state after performing an I/O system call if non-blocking I/O is used.

(2 marks)

(v) RAID (Redundant-Array-of-Independent-Disks) level 1 refers to non-redundant striping.

(2 marks)

Note: Question No. 4 continues on Page 7

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(vi) Buffering can be used to improve I/O efficiency for files that are being written and re-read rapidly.

(2 marks)

(b) Describe the problems of the continuous file allocation method and discuss how the index file allocation method overcomes these problems.

(4 marks)

(c) Assume a disk drive has 200 cylinders, numbered 0 to 199. The following requests, arranged according to arrival order, are in the disk queue: 120, 30, 130, 40, 180, and 190. The disk arm takes 2 msec per cylinder moved, and the service time (i.e., rotation latency plus data transfer time) per request is assumed to be 10 msec. How much time is needed to service all the requests in the queue for each of the following disk scheduling algorithms?

- First-Come-First-Served (FCFS)
- Shortest-Seek-Time-First (SSTF)
- SCAN (assuming the initial moving direction is outward)

In all cases, the disk head is initially at cylinder 60.

(6 marks)

END OF PAPER

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Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.